

From Dirty Data to Predictive Models: UCI Heart Disease Dataset Analysis

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Runnable Colab Notebook:

https:

[//drive.google.com/file/d/1Kx_0doPak-9xwlfJL8ozM420p-HrPHpr/view?usp=sharing](https://drive.google.com/file/d/1Kx_0doPak-9xwlfJL8ozM420p-HrPHpr/view?usp=sharing)

1 Introduction

In the medical domain, accurate and timely diagnosis is critical for effective patient care. Cardiovascular diseases remain a leading cause of global mortality, making the early detection of heart disease a significant objective for predictive modeling. The primary problem addressed in this report is the classification of patients based on their medical attributes to predict the presence or absence of heart disease. By building an end-to-end machine learning pipeline, this study aims to evaluate and compare the efficacy of generative and discriminative modeling approaches, specifically, Naive Bayes and Linear Regression, on a real-world clinical dataset.

The analysis is conducted using the UCI Heart Disease dataset, a standard benchmark in the machine learning community for binary classification tasks. The dataset comprises 303 patient records, divided into 242 training samples and 61 testing samples. It contains 13 clinical features, which include demographic information (such as age and sex), physiological measurements (such as resting blood pressure and serum cholesterol), and diagnostic test results (such as resting electrocardiographic results and maximum heart rate achieved). The target variable indicates the diagnosis of heart disease. In its original form, the UCI target ranges from 0 to 4; for this analysis, it was explicitly converted into a binary classification task where 0 represents the absence of disease and any value greater than 0 (1–4) indicates its presence. Due to the real-world nature of the data, the initial dataset contains challenges such as categorical variables that require encoding and potential structural inconsistencies. The UCI metadata indicates missing values in the `ca` and `thal` features, which were addressed in the cleaning phase through evidence-based imputation.

2 Data Cleaning

Real-world medical datasets are frequently hampered by missing values and noisy entries, which can degrade model performance or cause algorithms to fail. The initial phase of the pipeline focused on systematically identifying and resolving these data quality issues to ensure a robust foundation for feature engineering and modeling.

The first step involved standardizing missing value representations and quantifying their prevalence. While legacy versions of the dataset sometimes use “?” to denote missing data, the `ucimlrepo` standardizes these as `NaN`. An analysis of missingness revealed that only two features

contained null values: **ca** (number of major vessels colored by fluoroscopy) had 4 missing values (1.32%), and **thal** (thalassemia) had 2 missing values (0.66%).

To address these, a threshold-based decision framework was established: columns with greater than 50% missingness would be dropped, while those with under 10% missingness would be imputed. Because both features fell well below the 10% threshold, imputation was deemed the optimal strategy to preserve the overall sample size (303 records). A median imputation strategy was selected due to its robustness against potential outliers. Consequently, missing values in **ca** were filled with its median value of 0.0, and missing values in **thal** were filled with its median of 3.0.

In addition to missing value treatment, the cleaning pipeline incorporated a domain-specific outlier detection step to screen for biological impossibilities. Specifically, the data was checked for instances where serum cholesterol (**chol**) or resting blood pressure (**trestbps**) were recorded as zero or negative values. A review of the dataset distributions confirmed no such anomalies existed (minimum **chol** = 126; minimum **trestbps** = 94), meaning no rows required deletion for biological invalidity.

Table 1 summarizes the state of the dataset before and after the cleaning pipeline was applied. The final cleaned dataset retained all 303 original rows and 13 features with zero missing values, perfectly preparing the data for the subsequent transformation phase.

Table 1: Before and After Data Cleaning

Metric / Feature	Before Cleaning	After Cleaning
ca Missingness	4 NaN values (1.32%)	0 missing (filled with median 0.0)
thal Missingness	2 NaN values (0.66%)	0 missing (filled with median 3.0)
Biological Outliers	Unverified	Verified (none found)
Dataset Shape	303×13 (with missingness)	303×13 (complete dataset)

3 Feature Engineering

With a clean dataset established, the next phase of the pipeline involved engineering and transforming features to maximize their predictive signal and ensure compatibility with the downstream machine learning algorithms. The feature engineering pipeline was executed in five sequential steps: categorical encoding, feature construction, skewness transformation, data splitting, and feature scaling.

First, categorical variables identified in the UCI metadata were appropriately encoded. Binary features (**sex**, **fbs**, and **exang**) inherently possessed a 0/1 structure and were left unchanged. Multi-level categorical features with more than two unique values (**cp**, **restecg**, **slope**, and **thal**) required transformation to prevent the models from incorrectly interpreting them as ordinal. These were one-hot encoded. To avoid multicollinearity, often referred to as the dummy variable trap, which can cause numerical instability in linear models, the **drop_first=True** parameter was applied, successfully creating linearly independent boolean columns.

Second, domain-inspired interactions were constructed to capture physiological relationships that isolated features might miss. Two specific ratios were engineered, incorporating a small constant ($\epsilon = 10^{-6}$) to prevent zero-division errors:

1. **Cardiovascular Load Proxy (**chol_bp_ratio**):** Calculated as $\frac{\text{chol}}{\text{trestbps} + \epsilon}$, representing the relationship between serum cholesterol and resting blood pressure.
2. **Relative Exercise Capacity (**age_hr_ratio**):** Calculated as $\frac{\text{age}}{\text{thalach} + \epsilon}$, representing age divided by maximum heart rate (**thalach**), serving as a proxy for exercise capacity relative to age.

Third, continuous numerical features were evaluated for right-skewness, which can negatively impact algorithms assuming normal distributions. A threshold of a skewness strictly greater than 1.0 was established. For features meeting this criterion and containing no negative values (minimum ≥ 0), a logarithmic transformation, $\log(1 + x)$, was applied. This successfully normalized the distributions of `chol`, `oldpeak`, `ca`, and the newly constructed `chol_bp_ratio`. The `age_hr_ratio` exhibited a skewness below 1.0 and was consequently left unlogged.

Fourth, to ensure a rigorous evaluation and prevent data leakage, the dataset was split into training (80%) and testing (20%) sets. A stratified split with a fixed random seed (`random_state=42`) was utilized to preserve the target class balance across both subsets.

Finally, continuous features were standardized using Z-score normalization (mean of 0, variance of 1) via a `StandardScaler`. Crucially, the scaler was fitted exclusively on the training set and then used to transform both the training and testing sets. This scaling process is vital to ensure that features with larger magnitudes do not disproportionately dominate the objective functions, providing fair regularization for linear models.

4 Model Comparison

With the features engineered and scaled, the predictive modeling phase focused on comparing two distinct paradigms: a generative approach (Naive Bayes) and a discriminative approach (Linear Regression).

4.1 Training Setup

For the generative model, `BernoulliNB` was selected over `GaussianNB` because the categorical encoding and transformations resulted in predominantly binary features. To fully satisfy the Bernoulli distribution assumption, an explicit binarization step was applied: continuous features were binarized using the training-set median as a threshold (values above the median $\rightarrow 1$, else 0). These training thresholds were subsequently applied to the test set to avoid data leakage. Two variations were trained using Laplace smoothing parameters of $\alpha = 1.0$ and $\alpha = 0.01$ (avoiding $\alpha = 0$ to prevent zero-probability errors).

For the discriminative model, Linear Regression was adapted for binary classification. Because standard Ordinary Least Squares (OLS) outputs continuous values, a strict decision threshold of 0.5 was applied: predictions ≥ 0.5 were classified as 1 (disease present), and predictions < 0.5 as 0 (disease absent). For calculating the ROC-AUC, the continuous outputs were clipped to the $[0, 1]$ range. Alongside standard OLS, regularized variants—Ridge (L_2 penalty, $\alpha = 1.0$) and Lasso (L_1 penalty, $\alpha = 0.01$)—were also trained to evaluate if penalizing feature weights would improve generalization.

4.2 Evaluation Results

The models were evaluated on the 20% test set using Accuracy, Precision, Recall, F1-Score, and ROC-AUC. Table 2 summarizes the performance of all tested models.

Table 2: Test-Set Performance Metrics

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
BernoulliNB ($\alpha = 1.0$)	0.869	0.812	0.929	0.867	0.961
BernoulliNB ($\alpha = 0.01$)	0.836	0.800	0.857	0.828	0.947
Linear Regression (OLS)	0.902	0.893	0.893	0.893	0.942
Ridge (L_2 , $\alpha = 1.0$)	0.885	0.862	0.893	0.877	0.949
Lasso (L_1 , $\alpha = 0.01$)	0.885	0.862	0.893	0.877	0.948

Visualizations of model performance are provided below. Figure 1 displays the ROC curves for both model families, while Figures 2 and 3 illustrate the confusion matrices for Naive Bayes and Linear Regression, respectively.

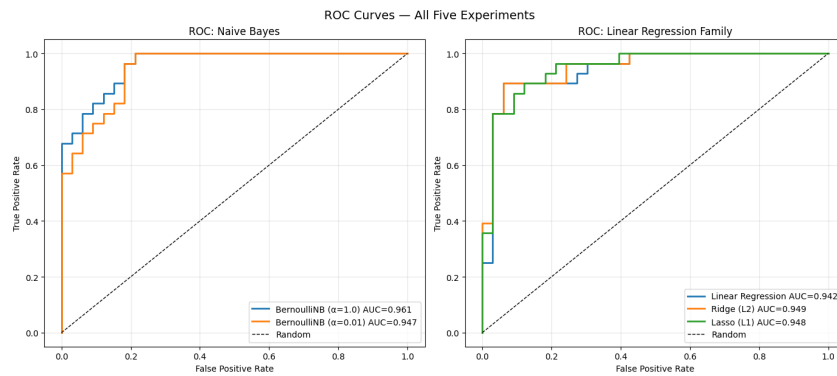


Figure 1: Receiver Operating Characteristic (ROC) Curves.

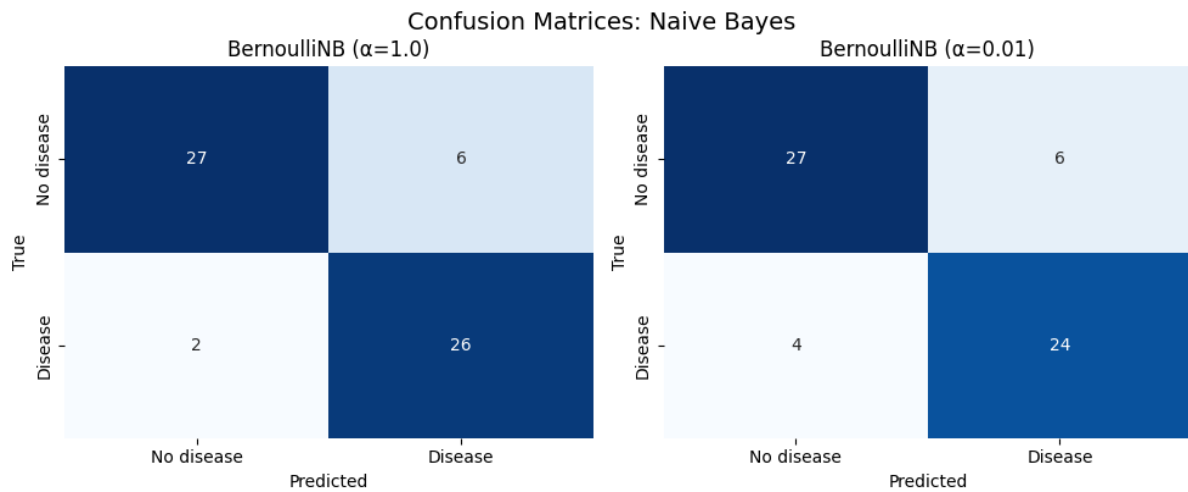


Figure 2: Confusion Matrix: BernoulliNB ($\alpha = 1.0$)

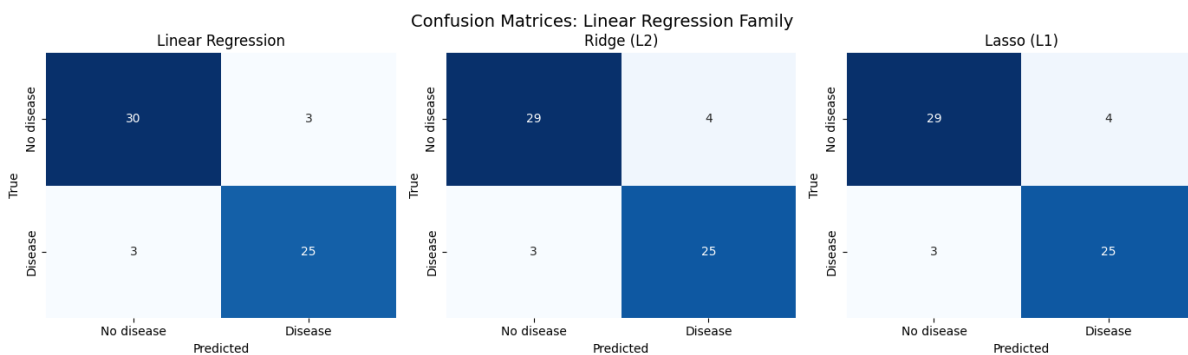


Figure 3: Confusion Matrix: Linear Regression (OLS)

5 Discussion

The results of the five experimental configurations highlight classical trade-offs in predictive modeling, deeply intertwined with the specific characteristics of the medical dataset used.

5.1 Dataset Characteristics and Model Suitability

The UCI Heart Disease dataset is relatively small, comprising 303 total samples partitioned into 242 training and 61 testing samples. This sample size is typical for the medical domain, where data collection is expensive and strictly regulated. The feature space is of moderate dimensionality, consisting of 13 original variables expanded to roughly 25 features after one-hot encoding and the construction of domain-driven interaction terms (such as `chol_bp_ratio`). The stratified split preserved a healthy class balance, avoiding the severe class imbalances often seen in diagnostics. In such environments, where the data is well-structured and features carry strong, known clinical linkages to the target, simpler models utilizing linear decision boundaries frequently rival or outperform more complex probabilistic models.

5.2 Analysis of Linear Regression Performance

Among the configurations tested, unregularized Linear Regression (OLS) achieved the highest overall Accuracy (90.2%) and F1-Score (89.3%). With approximately 25 features and 242 training samples, the roughly 10:1 observation-to-predictor ratio indicates the model is not severely overparameterized. When the underlying relationships are roughly linear and the features are highly informative, OLS can fit the data well without the need for regularization.

Interestingly, the introduction of L_1 (Lasso, $\alpha = 0.01$) and L_2 (Ridge, $\alpha = 1.0$) regularization slightly degraded performance metrics across the board. Rather than implying that regularization inherently dampens useful clinical signals, this outcome suggests a regularization bias: the unregularized feature weights did not exhibit enough variance to warrant the shrinkage penalties imposed by the chosen α hyperparameters. The shrinkage introduced slightly more bias than was optimal, reducing the overall fit compared to the OLS baseline.

Furthermore, Ridge and Lasso yielded identical discrete metrics (Accuracy, Precision, Recall, and F1-Score), with ROC-AUC differing only slightly (0.949 vs 0.948). While L_1 and L_2 penalties mathematically operate differently, with Lasso capable of driving coefficients to absolute zero, the identical performance in discrete predictions is a byproduct of the strict 0.5 classification threshold applied to the continuous outputs. Even if the raw predicted probabilities differed slightly between the two regularized models, the values fell on the exact same side of the decision boundary for every single patient in the test set. Additionally, medical predictors (e.g., age, cholesterol, blood pressure) often exhibit collinearity; depending on the penalty size, both regularization strategies likely converged to functionally equivalent decision boundaries.

5.3 Clinical Trade-offs: Accuracy vs. Recall

Despite OLS achieving the highest accuracy, medical diagnostic tasks demand a careful weighting of false negatives against false positives. In predicting heart disease, missing a positive case (False Negative) can delay critical treatment, making it generally much more detrimental than a false alarm (False Positive). Therefore, Recall is arguably the most critical metric. The generative model, BernoulliNB ($\alpha = 1.0$), achieved the highest Recall (92.9%) and the highest ROC-AUC (0.961). By modeling the joint probability distribution of the heavily binarized feature space, Naive Bayes demonstrated superior capability in correctly flagging patients who actually suffer from heart disease, albeit at the cost of lower precision.

5.4 Visual Analysis of Model Performance

The theoretical trade-offs between the generative and discriminative approaches are clearly reflected in the evaluation visualizations. As illustrated in the confusion matrices (Figures 2 and 3), Linear Regression (OLS) exhibits a highly balanced error distribution, which translates to its identical precision and recall scores (89.3%). Conversely, the confusion matrix for BernoulliNB visibly demonstrates a shift in the error profile: it successfully minimizes False Negatives at the expense of an increase in False Positives. This visual confirmation aligns directly with its utility as a conservative diagnostic tool where missing a positive diagnosis is heavily penalized.

Furthermore, the Receiver Operating Characteristic (ROC) curves (Figure 1) provide deeper insight into the models' discriminative capabilities across all possible classification thresholds. The BernoulliNB model's curve visibly hugs the top-left corner more closely than the Linear Regression family, resulting in the highest overall AUC (0.961). This indicates that, independent of the strict 0.5 threshold used for the discrete metrics, the generative model provides a superior and more robust probabilistic ranking of the patients' actual disease risk.

5.5 Limitations

A critical limitation of this study is the high variance associated with the small test set size. With only 61 test samples, a difference between 90.2% (OLS) and 88.5% (Ridge/Lasso) equates to exactly one patient being classified differently. Consequently, we cannot definitively conclude that OLS consistently generalizes better than regularized models on this dataset without implementing k-fold cross-validation. Finally, while OLS performed admirably, utilizing it for binary classification constitutes a Linear Probability Model, which violates fundamental regression assumptions (e.g., heteroscedastic residuals and unbounded outputs). In a rigorous production environment, Logistic Regression would be the mathematically appropriate alternative to establish a linear decision boundary.

6 AI Tool Usage Disclosure

- **AI Tools Used:** Gemini Pro and Cursor.
- **Contribution of AI Tools:** Gemini Pro was utilized as a conceptual tutor to guide me through the assignment requirements without providing direct code answers. Cursor was used within the IDE for code autocompletion, rapid syntax debugging, and pulling inline documentation for `scikit-learn` and `pandas` functions.
- **Personal Contribution:** The core analytical decisions were entirely my own. This includes formulating the framework for missing value imputation, designing the domain-specific feature engineering logic (such as creating the `chol_bp_ratio` and selectively applying log transformations based on skewness), choosing the evaluation metrics, and interpreting the clinical trade-offs between OLS and Naive Bayes in the discussion.